PROBLEMS

Problem (1) A local city that operates automobile parking facilities is evaluating a proposal that it will build and operate a structure for parking in the city's downtown area. Three designs for the facility to be built on available sites have been identified in the following table.

	Design A	Design B	Design C
Cost of Site	$240	$180	$200
Cost of Building	$2,200	$700	$1,400
Annual Fee Collection	$830	$750	$600
Annual Maintenance Cost	$410	$360	$310
Service Life (Years)	30	30	30

If the city's interest rate 10%,

a) What would be the benefit-cost (BC) ratio for each design?

Design A:
B = \$830 (P/A, 10%, 30) = \$7,824.41
$I + C'$ = \$240 + \$2,200 + \$410 (P/A, 10%, 30) = \$2,440 + \$3,865.07 = \$6,305.07
$BC(10\%)_A$ = \$7,824.41 / \$6,305.07 = 1.24 > 1

Design B:
B = \$750 (P/A, 10%, 30) = \$7,070.25
$I + C'$ = \$180 + \$700 + \$360 (P/A, 10%, 30) = \$880 + \$3,393.72 = \$4,273.72
$BC(10\%)_B$ = \$7,070.25 / \$4,273.72 = 1.65 > 1

Design C:
B = \$600 (P/A, 10%, 30) = \$5,656.20
$I + C'$ = \$200 + \$1,400 + \$310 (P/A, 10%, 30) = \$1,600 + \$2,922.37 = \$4,522.37
$BC(10\%)_C$ = \$5,656.20 / \$4,522.37 = 1.25 > 1

b) Which design would be selected on the basis of the BC(i) criterion?

$BC(10\%)_{C-B}$ = (\$5,656.20 - \$7,070.25) / (\$4,522.37 - \$4,273.72) = -5.69 < 1
Select B

$BC(10\%)_{A-B}$ = (\$7,824.41 - \$7,070.25) / (\$6,305.07 - \$4,273.72) = 0.37 < 1
Select B

Problem (2) A corporation is trying to decide whether to buy a patent for a product designed by another company. The decision to buy will mean an investment of $7,000,000 and the demand is unknown. If the demand is low, the company expects a return of $1,100,000 each year for 3 years. If the demand is moderate, the return will be $2,400,000 each year for 4 years. A high demand means a return of $4,800,000 each year for 4 years.

It is estimated that the probability of a low demand is 25%, of a moderate demand is 30% and a high demand is 45%. The company's risk-free interest rate is 14%. Calculate the expected present worth of the patent.

$PW(14\%)_{low}$ = -$7,000,000 + $1,100,000 (*P/A*, 14%, 3) = -$4,446,240

$PW(14\%)_{moderate}$ = -$7,000,000 + $2,400,000 (*P/A*, 14%, 4) = -$7,120

$PW(14\%)_{high}$ = -$7,000,000 + $4,800,000 (*P/A*, 14%, 4) = $6,985,760

E[PW(14%)] = -$4,446,240 (25%) + -$7,120 (30%) + $6,985,760 (45%) = $2,029,896

Problem (3) Consider a 5-year MACRS asset purchased at $80,000. The applicable salvage values would be $40,000 in year 3 and $30,000 in year 5. Compute the gain or loss amounts when the asset is disposed of in

a) Year 3 b) Year 5

Use the depreciation rates for MACRS class below for personal property with half-year conversation.

Year	Class Depreciation Rate	3 200% DB	5 200% DB
1		33.33	20.00
2		44.45	32.00
3		14.81*	19.20
4		7.41	11.52*
5			11.52
6			5.76

a) Disposed in Year 3:

Allowed depreciation = $80,000 (0.20 + 0.32 + 0.192/2) = $80,000 (0.616) = $49,280

Book value = $80,000 - $49,280 = $30,720

Gains = Salvage value – Book value = $40,000 - $30,720 = $9,280

b) Disposed in Year 5:

Allowed depreciation = $80,000 (0.20 + 0.32 + 0.192 + 0.1152 + 0.1152/2)

= $80,000 (0.8848) = $70,784

Book value = $80,000 - $70,784 = $9,216

Gains = Salvage value – Book value = $30,000 - $9,216 = $20,784

Problem (4) Neo Limousine Service uses the units of production method in computing depreciation on its limos. Each limo costs $36,000, is to be driven 200,000 miles in total, and is expected to have a salvage value of $4,000. Limo #1 was driven 20,000 miles in year 1 and 25,000 miles in year 2. Determine the depreciation for each year and the book value at the end of year 2.

D_1 = (20,000 / 200,000) ($36,000 - $4,000) = $3,200
D_2 = (25,000 / 200,000) ($36,000 - $4,000) = $4,000

Book Value = $36,000 - $3,200 - $4,000 = $28,800

Single Series Compound Amount Factor *(F/P, i% N)* : $F = P(1 + i)^N$

Single Series Present Worth Factor *(P/F, i%, N)* : $P = F(1 + i) -^N$

Equal Payment Series Compound Amount Factor *(F/A, i%, N)* : $F = A \left[\frac{(1 + i)^N - 1}{i}\right]$

Equal Payment Series Sinking Fund Factor *(A/F, i%, N)* : $A = F \left[\frac{i}{(1 + i)^N - 1}\right]$

Equal Payment Series Present Worth Factor *(P/A, i%, N)* : $P = A \left[\frac{(1 + i)^N - 1}{i(1 + i)^N}\right]$

Equal Payment Series Capital Recovery Factor *(A/P, i%, N)* : $A = P \left[\frac{i(1 + i)^N}{(1 + i)^N - 1}\right]$

Uniform Gradient Series Conversion Factor *(A/G, i%, N)* : $A = G \left[\frac{(1 + i)^N - iN - 1}{i(1 + i)^N - 1}\right]$

Uniform Gradient Present Worth Factor *(P/G, i%, N)* : $P = G \left[\frac{(1 + i)^N - iN - 1}{i2(1 + i)^N}\right]$

Geometric Gradient Series Present Worth Factor *(P/A,g%,i%,N)*: $P = A \left[\frac{1 - (1+g)^N (1+i) - N}{i - g}\right]$

N	Single Payment: Compound Amount Factor F/P	Single Payment: Present Worth Factor P/F	Uniform Payment Series: Capital Recovery Factor A/P	Uniform Payment Series: Present Worth Factor P/A	Uniform Payment Series: Sinking Fund Factor A/F	Uniform Payment Series: Compound Amount Factor F/A	Arithmetic Gradient: Present Worth Factor P/G	Arithmetic Gradient: Uniform Payment Factor A/G	N
1	1.100	.9091	1.1000	.909	1.0000	1.000	0	0	1
2	1.210	.8264	.5762	1.736	.4762	2.100	.826	.476	2
3	1.331	.7513	.4021	2.487	.3021	3.310	2.329	.937	3
4	1.464	.6830	.3155	3.170	.2155	4.641	4.378	1.381	4
5	1.611	.6209	.2638	3.791	.1638	6.105	6.862	1.810	5
6	1.772	.5645	.2296	4.355	.1296	7.716	9.684	2.224	6
7	1.949	.5132	.2054	4.868	.1054	9.487	12.763	2.622	7
8	2.144	.4665	.1874	5.335	.0874	11.436	16.029	3.004	8
9	2.358	.4241	.1736	5.759	.0736	13.579	19.421	3.372	9
10	2.594	.3855	.1627	6.145	.0627	15.937	22.891	3.725	10
11	2.853	.3505	.1540	6.495	.0540	18.531	26.396	4.064	11
12	3.138	.3186	.1468	6.814	.0468	21.384	29.901	4.388	12
13	3.452	.2897	.1408	7.103	.0408	24.523	33.377	4.699	13
14	3.797	.2633	.1357	7.367	.0357	27.975	36.800	4.996	14
15	4.177	.2394	.1315	7.606	.0315	31.772	40.152	5.279	15
16	4.595	.2176	.1278	7.824	.0278	35.950	43.416	5.549	16
17	5.054	.1978	.1247	8.022	.0247	40.545	46.582	5.807	17
18	5.560	.1799	.1219	8.201	.0219	45.599	49.640	6.053	18
19	6.116	.1635	.1195	8.365	.0195	51.159	52.583	6.286	19
20	6.727	.1486	.1175	8.514	.0175	57.275	55.407	6.508	20
21	7.400	.1351	.1156	8.649	.0156	64.002	58.110	6.719	21
22	8.140	.1228	.1140	8.772	.0140	71.403	60.689	6.919	22
23	8.954	.1117	.1126	8.883	.0126	79.543	63.146	7.108	23
24	9.850	.1015	.1113	8.985	.0113	88.497	65.481	7.288	24
25	10.835	.0923	.1102	9.077	.0102	98.347	67.696	7.458	25
26	11.918	.0839	.1092	9.161	.00916	109.2	69.794	7.619	26
27	13.110	.0763	.1083	9.237	.00826	121.1	71.777	7.770	27
28	14.421	.0693	.1075	9.307	.00745	134.2	73.650	7.914	28
29	15.863	.0630	.1067	9.370	.00673	148.6	75.415	8.049	29
30	17.449	.0573	.1061	9.427	.00608	164.5	77.077	8.176	30
31	19.194	.0521	.1055	9.479	.00550	181.9	78.640	8.296	31
32	21.114	.0474	.1050	9.526	.00497	201.1	80.108	8.409	32
33	23.225	.0431	.1045	9.569	.00450	222.3	81.486	8.515	33
34	25.548	.0391	.1041	9.609	.00407	245.5	82.777	8.615	34
35	28.102	.0356	.1037	9.644	.00369	271.0	83.987	8.709	35
40	45.259	.0221	.1023	9.779	.00226	442.6	88.953	9.096	40
45	72.890	.0137	.1014	9.863	.00139	718.9	92.454	9.374	45
50	117.4	.00852	.1009	9.915	.00086	1164	94.889	9.570	50
55	189.1	.00529	.1005	9.947	.00053	1881	96.562	9.708	55
60	304.5	.00328	.1003	9.967	.00033	3035	97.701	9.802	60
65	490.4	.00204	.1002	9.980	.00020	4894	98.471	9.867	65
70	789.7	.00127	.1001	9.987	.00013	7887	98.987	9.911	70
75	1272	.00079	.1001	9.992	.000079	12709	99.332	9.941	75
80	2048	.00049	.1000	9.995	.000049	20474	99.561	9.961	80
85	3299	.00030	.1000	9.997	.000030	32980	99.712	9.974	85
90	5313	.00019	.1000	9.998	.000019	53120	99.812	9.983	90
95	8557	.00012	.1000	9.999	.000012	85557	99.877	9.989	95
100	13781	.00007	.1000	9.999	.000007	137796	99.920	9.993	100
∞	∞	0	.10	10	0	∞	100	10	∞